

COLDPOD · DESIGN DECISION RECORD

ColdPod recommendations accepted

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-26	Amish Chadha	Budget top-up to \$310 decided by Amish (N3); R15 met
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all recommendations in CPD-DDR-001 and docs/REVIEW.md, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item with a recommendation is decided by Amish; the item without a recommendation, and three new findings from the recalculation, stay open.

■ Context

CPD-DDR-001 decided the eleven TRL 2 review items (D1 to D11) and left two groups open: O1, the co-design partner, with no recommendation, and O2, the new proposals from the TRL 3 session of `docs/REVIEW.md` (items 2 to 8 in its list "Proposed, awaiting Amish"), each of which carried a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore **decided by Amish, 2026-09-25: go with recommendation**. Where a recommendation offered several options, the recommended option is the decision. Items with no recommendation stay "Proposed, awaiting Amish". TRL 4 remains on hold by Amish's instruction, so decisions that need building, testing, weighing or purchasing are recorded but not carried out.

D1 to D11 in CPD-DDR-001 already read "Decided by Amish, 2026-09-25: go with recommendation" and are unchanged.

■ Decision

Table 1. Items decided by Amish on 2026-09-25 and what changed in the repo. Item numbers follow the TRL 3 session of `docs/REVIEW.md`.

#	Decision	What changed in the repo
T2	Lid pack cold path: option (a), a 1.5 mm aluminium plate under the lid PCM pack that seats on the evaporator can rim when the lid closes (about \$5, 0.12 kg)	Plate added to <code>cad/src/model.py</code> (<code>lid_plate_t</code> , exported with the lid), BOM line 2 (\$8 to \$12), drawing CPD-DWG-001 Rev P2, media. CPD-CAL-001 v0.2: the lid pack now freezes through the plate and stays frozen near 2.8 °C in powered hold at 43 °C, so R1 and R7 move from at risk to met; hold times pooled (R4 13.0 to 14.0 h, R5 26.5 to 28.4 h, R6 16.5 to 17.5 h). Refreeze of all the PCM is now 8.4 h (jacket alone was 5.5 h), so R8 stays not met, for a new reason (see Table 2). Checking the gasket seal with the plate is a test on a built lid, on hold with TRL 4
T3	Freeze fault: a second hardware cut-out on the liner, opening at 3 °C, in series with the cold-block cut-out (about \$3)	Cut-out added to the model and to BOM line 14 (\$10 to \$13); wiring named in BOM line 12. CPD-CAL-001 v0.2 section I: liner settles near 2.0 °C after a stuck-on fault (was about −4 °C); R2 moves from not met to met on paper; R2 target text in CPD-REQ-001 v0.4 names both cut-outs
T4	Mass: option (a), thinner printed parts (shell 4 to 3 mm, lid cap 8 to 5 mm, end housings 3 to 2 mm)	<code>shell_t</code> , <code>lid_cap_t</code> and <code>end_wall</code> changed in <code>cad/src/model.py</code> ; BOM lines 1 (\$18 to \$17) and 10 (\$8 to \$6); drawing notes. Mass 5.60 to 5.53 kg with the plate; R12 still not met, by 0.03 kg (the review estimated about 5.51 kg). Confirming by weighing is TRL 4, decided but on hold
T5	Evaporator can and loop thermosiphon confirmed as sized (1.0 mm aluminium can, 8 mm copper loop), with the mechanical disconnect (D2 fallback) kept ready	Status wording in CPD-PRC-001 v0.4 and CPD-CAL-001 v0.2; no geometry change. Charging a loop is build work, on hold
T6	Peltier class and drive confirmed: TEC1-12703 class, buck driver stage with LC filter for smooth DC, heat sink 0.50 K/W or better with the fan	Status wording only; already in the model and BOM lines 8, 9 and 12. The driver stage becomes buck-boost under T7
T7	Vehicle input headroom: option (a), a buck-boost Peltier driver	BOM line 12 (\$22 to \$26); R7 target text in CPD-REQ-001 v0.4 names the buck-boost driver for the 10 to 15 V input; CPD-CAL-001 v0.2 E9b. R7 met on paper
T8	Budget margin: <code>budget_usd</code> stays at \$300; get a VIP quote before any purchase decision	<code>project.yaml</code> unchanged at <code>budget_usd: 300</code> . Requesting supplier quotes belongs with purchasing, which is TRL 4 work: decided but on hold

Other effects:

- **Budget.** `budget_usd` stays at 300 (T8). The BOM moves from \$295 to \$303 (plate +\$5, liner cut-out +\$3, buck-boost +\$4, thinner prints −\$3, lid cap filament −\$1), so R15 moves from met to not met by \$3. This is a finding, not a decision; see Table 2.
- **Pitch and problem.** No recommendation changed them; `project.yaml` and `README.md` keep the same lines.
- **Size.** The body is 2 mm shorter and narrower and 2.5 mm lower at the lid top; overall 366 x 212 x 207 mm (was 368 x 214 x 207 mm).
- **Controlled documents changed:** CPD-DDR-001 v0.2, CPD-PRB-001 v0.4, CPD-PRC-001 v0.4, CPD-REQ-001 v0.4, CPD-CAL-001 v0.2; drawing CPD-DWG-001 Rev P2.
- `trl` and `trl_target` stay at 3.

■ Items still open

Table 2. Proposed, awaiting Amish.

#	Item	Status
O1	Co-design partner: an immunization program, a diabetes association or a humanitarian logistics group	Proposed, awaiting Amish. No recommendation was made; partners are picked per area later
N1	R8 refreeze: 8.4 h for all the PCM against 8 h, now that the lid pack is frozen actively. Giving the module 30 W and the battery 7 W only reaches 8.2 h (CPD-CAL-001, G10). Options: (a) restate R8 as 9 h for all the PCM; (b) accept the miss; (c) a thinner lid pack, trading passive hold	New finding from CPD-CAL-001 v0.2. Proposed, awaiting Amish; suggestion (a)
N2	R12 mass: 5.53 kg against 5.5 kg. Options: (a) keep the target and settle it by weighing when TRL 4 resumes; (b) relax to 5.6 kg; (c) a lighter strap and handle	New finding. Proposed, awaiting Amish; suggestion (a)
N3	R15 cost: \$303 against the \$300 budget. Options: (a) raise <code>budget_usd</code> to \$310; (b) keep \$300 and wait for the VIP quote (on hold with TRL 4); (c) keep \$300 and drop the buck-boost stage for an 11 to 15 V vehicle input (saves about \$4, reopens R7)	New finding. Budget top-up to \$310: decided by Amish, 2026-09-26. <code>budget_usd</code> set to 310; R15 met (\$303 against \$310, CPD-CAL-001 v0.3)

■ Budget top-up, 2026-09-26

Budget top-up to \$310: decided by Amish, 2026-09-26 ("I am ok with the budget top ups"). This closes N3 with option (a). `project.yaml` now carries `budget_usd: 310`, R15 in CPD-REQ-001 v0.5 reads \$310, and CPD-CAL-001 v0.3 rechecks the \$303 BOM against it: met, with \$7 of margin.

■ Cross-repo actions

None. ColdPod shares no part or interface with another portfolio repo: it does not use the SwapCell pack (CPD-DDR-001), and no recommendation named another repo.

■ Consequences

- Requirement status (CPD-CAL-001 v0.3, after the budget top-up): 2 not met (R8, R12, each by a small margin), 3 at risk (R4, R5, R6), 10 met, 2 not verifiable at TRL 3 (R10, R16). In v0.2, before the top-up, R15 was also not met. Before: 3 not met (R2, R8, R12), 5 at risk, 7 met, 2 not verifiable.
- The design has no remaining freeze path on paper and no lid pack that runs warm in long hot holds.
- TRL 4 work (weighing, gasket and plate seat checks, loop charging, VIP quotes and purchasing, climate chamber tests) stays on hold.